CAPSTONE PROJECT REPORT

SAP ERP Implementation Blueprint

Covering FI, MM & SD Modules with End-to-End Process Flows

Field	Details
Name	A***** R***
Roll Number	2305*****
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Company: TechNova Manufacturing Pvt. Ltd. | Domain: Electronics Manufacturing | ERP: SAP ECC 6.0 / S/4HANA

1. Introduction

In today's dynamic manufacturing environment, implementing a robust Enterprise Resource Planning (ERP) system is not merely a technological decision — it is a strategic imperative. Organizations that leverage SAP ERP are better positioned to control costs, streamline operations, and achieve real-time visibility across their supply chain, financial, and sales functions.

SAP (Systems, Applications and Products in Data Processing) is the world's leading ERP platform, trusted by over 440,000 organizations globally. SAP integrates core business functions — including Finance (FI), Materials Management (MM), and Sales & Distribution (SD) — into a unified system, enabling automated workflows, cross-functional transparency, and real-time data access.

This capstone project presents a comprehensive SAP ERP implementation blueprint for TechNova Manufacturing Pvt. Ltd., a fictitious mid-sized Indian electronics manufacturer headquartered in Bengaluru. The blueprint covers the complete organizational structure, module-wise configuration, master data design, and end-to-end business process documentation across the Procure-to-Pay (P2P), Order-to-Cash (O2C), and Record-to-Report (R2R) cycles.

2. Objective

The primary objectives of this capstone project are:

- To design the complete SAP enterprise structure for TechNova Manufacturing covering FI, MM, and SD modules.
- To document step-by-step configuration checklists with SAP transaction codes (T-Codes) for each module.
- To create realistic master data including 35 GL accounts, 15 vendors, 20 materials, and 15 customers.
- To simulate end-to-end business processes: Procure-to-Pay (P2P), Order-to-Cash (O2C), and Record-to-Report (R2R).
- To document cross-module integration between FI, MM, and SD with automatic accounting entries.
- To demonstrate practical SAP functional knowledge applicable to real-world implementation scenarios.

3. Company Overview

TechNova Manufacturing Pvt. Ltd.

TechNova Manufacturing Pvt. Ltd. is a mid-sized Indian electronics components manufacturer headquartered in Bengaluru, Karnataka. The company procures raw materials, manufactures electronic sub-assemblies, and sells finished products to B2B clients across India and exports to Southeast Asia. The company has an annual turnover of approximately ₹120 Crore and employs around 850 people.

Industry	Electronics Manufacturing
Headquarters	Bengaluru, Karnataka, India
Annual Turnover	₹120 Crore (approx.)
Employees	~850
ERP System	SAP ECC 6.0 (S/4HANA Ready Blueprint)
Modules Covered	SAP FI, SAP MM, SAP SD
Company Code	TN01
Currency	INR (Indian Rupee)
Fiscal Year	V3 — April to March (Indian FY)

4. SAP Organizational Structure

Before any SAP configuration can begin, the organizational structure must be defined. This hierarchy maps the company's legal, logistical, and commercial structures into SAP. The structure for TechNova spans all three modules:

4.1 Financial Accounting (FI) Structure

Org. Unit	Key / ID	Description
Client	100	TechNova Group
Company	TNOV	TechNova Group Holding
Company Code	TN01	TechNova Manufacturing Pvt. Ltd.
Chart of Accounts	TNCA	TechNova Chart of Accounts (6-digit GL)
Fiscal Year	V3	April to March — 12 periods + 4 special

Org. Unit	Key / ID	Description
Business Area	TN10	Electronics Manufacturing
Business Area	TN20	Trading & Procurement
Business Area	TN30	Export Division
Controlling Area	TN01	TechNova Controlling Area

4.2 Materials Management (MM) Structure

Org. Unit	Key / ID	Description
Plant	TN01	TechNova Main Plant — Bengaluru
Plant	TN02	TechNova Warehouse — Hosur
Storage Location	SL01	Raw Material Store
Storage Location	SL02	Work-In-Progress (WIP) Store
Storage Location	SL03	Finished Goods Store
Storage Location	SL04	Rejection / Quarantine Store
Purchasing Org.	TN01	TechNova Central Purchasing
Purchasing Group	TNG1	Raw Material Buyers
Purchasing Group	TNG2	Capital Goods Buyers
Purchasing Group	TNG3	Services & MRO

4.3 Sales & Distribution (SD) Structure

Org. Unit	Key / ID	Description
Sales Organization	TN01	TechNova India Sales
Distribution Channel	10	Direct Sales
Distribution Channel	20	Dealer / Distributor
Distribution Channel	30	Export
Division	01	Electronic Components
Division	02	Sub-Assemblies
Division	03	Finished Products
Shipping Point	TNSP	TechNova Dispatch Dock — Plant TN01

5. Module Configuration Overview

The configuration of TechNova's SAP system follows a structured approach across three phases for each module. Below is a summary of the key configuration activities with their corresponding SAP transaction codes.

5.1 SAP FI — Financial Accounting Configuration

Phase	Configuration Activity	T-Code
Enterprise Structure	Define Company, Company Code, Assign to Company	OX15, OX02, OX16
Chart of Accounts	Create CoA (TNCA), Define Account Groups, Assign	OB13, OBD4, OB62
Fiscal Year	Define Fiscal Year Variant V3 (Apr–Mar), Assign	OB29, OB37
Document Types	Configure document types (SA, KR, DR, WA, RE)	OBA7, FBN1
Posting Keys	Define posting keys 40/50 (GL), 01/31 (AR/AP)	OB41
Accounts Payable	Vendor Account Groups, Reconciliation Account, Terms	OBD3, FBZP
Accounts Receivable	Customer Account Groups, Recon. Account, Dunning	OBD2, FBMP
Tax / GST	Assign TAXIN procedure, configure GST codes	OBYZ, OBBG

5.2 SAP MM — Materials Management Configuration

Phase	Configuration Activity	T-Code
Enterprise Structure	Define Plant, Storage Locations, Pur. Org, Assign	OX10, OX09, OX08
Material Master	Material Types (ROH/HALB/FERT/HIBE), Groups, UoM	OMS2, OMSF
Valuation	Valuation Level (Plant), Classes, OBYC Account Det.	OX14, OMSK, OBYC
Purchasing	Pur. Groups, PO Doc Types, Number Ranges, Release	OME4, OMH6, OMH7
Inventory Mgmt.	Movement Types overview, ABC Classification, Counts	OMJJ, MI01
Invoice Verification	LIV Tolerances, Payment Block Reasons, GR-Based IV	OMR6

5.3 SAP SD — Sales & Distribution Configuration

Phase	Configuration Activity	T-Code
Enterprise Structure	Sales Org, Distribution Channel, Division, Sales Area	OVX5, OVX1, OVXG
Customer Master	Account Groups (DOME/EXPG/DIST), Number Ranges	OVT0, VN01
Sales Documents	Document types (IN/QT/OR/RE), Item Categories	VOV8, VOV7
Pricing	Condition Types (PR00, K004, MWST), Pricing Procedure	V/06, V/08, OVKK
Delivery	Delivery Types, Shipping Conditions, Goods Issue	OVLK
Billing	Billing Types (F2/G2), Revenue Account Determination	VOFA, VKOA
Credit Management	Credit Control Area, Credit Limits, Static Check	OB45, FD32, OVA8

6. Master Data

Master data is the foundation of any SAP implementation. The following master data has been designed for TechNova Manufacturing across all three modules.

6.1 Chart of Accounts — Key GL Accounts (TNCA)

GL Account	Account Description	Type	Account Group
100200	Bank Account — SBI Current	Balance Sheet	ASST
100300	Accounts Receivable Control	Balance Sheet	ASST (Recon.)
130100	Raw Material Inventory	Balance Sheet	ASST
130300	Finished Goods Inventory	Balance Sheet	ASST
200100	Accounts Payable Control	Balance Sheet	LIAB (Recon.)
200200	GR/IR Clearing Account	Balance Sheet	LIAB
210300	GST Output IGST Payable	Balance Sheet	LIAB
400100	Domestic Sales Revenue	P&L	REVN
500100	Cost of Goods Sold — Material	P&L	COGS
560100	Salaries & Wages	P&L	EXPN

The complete chart of accounts (35 accounts) is available in the file: 02_FI_Configuration/chart_of_accounts.xlsx

6.2 Vendor Master — Summary

Vendor No.	Vendor Name	Account Group	Payment Terms
V1001	Arya Electronics Pvt. Ltd.	DOME (Domestic)	NET30
V1002	Bharat Components Ltd.	DOME (Domestic)	NET45
V1003	Circa Semiconductors India	DOME (Domestic)	2/10 NET30
V2001	Microchip Technology Inc.	FORG (Foreign)	NET45
V2002	Shenzhen ElecTech Co. Ltd.	FORG (Foreign)	NET60
V3001	National Facility Services	SERV (Service)	NET30

Full vendor master data (15 vendors) is available in: 03_MM_Configuration/vendor_master_data.xlsx

6.3 Material Master — Summary

Material No.	Description	Type	Std. Price (₹)
TN-RM-001	PCB Bare Board — 4 Layer	ROH (Raw Material)	₹145
TN-RM-006	IC Microcontroller ATmega328	ROH (Raw Material)	₹320
TN-SF-001	Motor Control PCB Assembly	HALB (Semi-Finished)	₹1,250
TN-FG-001	Industrial IoT Gateway — Model A	FERT (Finished Good)	₹8,500
TN-FG-002	Motor Drive Controller 5kW	FERT (Finished Good)	₹12,500
TN-MR-001	Soldering Iron — 60W	HIBE (MRO)	₹850

Full material master data (20 materials) is available in: 03_MM_Configuration/material_master_data.xlsx

6.4 Customer Master — Summary

Customer No.	Customer Name	Account Group	Credit Limit (₹)
C10001	Aditya Automation Pvt. Ltd.	DOME (Domestic)	₹25,00,000
C10003	Crest Electricals Ltd.	DOME (Domestic)	₹50,00,000
C20001	Apex Automation Singapore	EXPG (Export)	₹50,00,000
C20002	Bali Electrik Indonesia	EXPG (Export)	₹35,00,000
C30001	TechNova Dealers — North Zone	DIST (Distributor)	₹60,00,000

Full customer master data (15 customers) is available in: 04_SD_Configuration/customer_master_data.xlsx

7. Business Process Architecture

TechNova's SAP system supports three core end-to-end business processes. Each process creates a chain of linked documents ensuring complete traceability and auditability. SAP's three-way matching mechanism (PO + GR + Invoice) and document flow (Sales Order → Delivery → Billing) are key controls that ensure accuracy and prevent financial errors.

7.1 Procure-to-Pay (P2P) — SAP MM + FI

Need Identified → Purchase Requisition (PR) → PR Approval → Purchase Order (PO) → PO Approval → Goods Receipt (GR) → Invoice (MIRO) → Payment (F110)

P2P Stage	SAP Transaction	Description
1. Purchase Requisition	ME51N	Internal request to procure material; specifies material, qty, date, plant.
2. PR Release	ME54N	Manager approves PR before it can be converted to a PO.
3. Purchase Order	ME21N	Legal document sent to vendor committing to purchase at agreed price.
4. PO Release	ME29N	Head of Procurement approves PO for values above ₹50,000.
5. Goods Receipt	MIGO	Physical receipt of goods posted; stock increases, GR/IR account credited.
6. Invoice Verification	MIRO	3-way match (PO + GR + Invoice); vendor liability posted in FI.

P2P Stage	SAP Transaction	Description
7. Payment Run	F110	Automated payment run; clears vendor open item; bank account debited.

Sample FI Posting at Goods Receipt (Movement Type 101):

Account	GL No.	Debit / Credit	Amount
Raw Material Inventory	130100	Debit (Dr.)	₹29,000
GR/IR Clearing Account	200200	Credit (Cr.)	₹29,000

7.2 Order-to-Cash (O2C) — SAP SD + FI

Customer Inquiry → Quotation → Sales Order (SO) → Credit Check → Outbound Delivery → Post Goods Issue (PGI) → Billing → Payment Receipt

O2C Stage	SAP Transaction	Description
1. Quotation	VA21	Formal price offer to customer with validity period and pricing conditions.
2. Sales Order	VA01	Binding commitment; triggers availability check, credit check, and pricing.
3. Credit Check	FD32	Automatic check against customer credit limit; hard block if exceeded.
4. Outbound Delivery	VL01N	Delivery document created; triggers warehouse picking from SL03.
5. Post Goods Issue	VL02N	Goods dispatched; stock reduced (Mvt 601); COGS posted automatically.
6. Billing	VF01	Tax invoice generated; AR debited, revenue and GST output credited.
7. Payment Receipt	F-28	Customer payment posted; AR open item cleared against bank account.

Sample FI Posting at Billing (Billing Document F2 — ₹1,00,300 invoice):

Account	GL No.	Debit / Credit	Amount
Accounts Receivable Control	100300	Debit (Dr.)	₹1,00,300
Domestic Sales Revenue	400100	Credit (Cr.)	₹85,000
GST Output IGST Payable	210300	Credit (Cr.)	₹15,300

7.3 Record-to-Report (R2R) — SAP FI

The R2R cycle covers day-to-day GL accounting through month-end close to financial statement generation. Key month-end activities for TechNova are:

Activity	T-Code	Description
AP / AR Reconciliation	S_ALR_87012082	Reconcile subledger balances with GL control accounts.
Post Accruals & Provisions	FB50	Accrue salary, utilities, warranty provisions at month-end.
Depreciation Run	AFAB	Post monthly depreciation for all fixed assets.
GR/IR Clearing	F.13 / MR11	Clear matched GR/IR items; investigate aged items.
FX Revaluation	FAGL_FC_VAL	Revalue open foreign currency items at closing rate.
GST Reconciliation	Manual	Match input/output GST with GSTR-1 and GSTR-2B data.
Close Posting Period	OB52	Lock current period; prevent backdated postings.
Financial Statements	S_ALR_87012284	Generate Balance Sheet and P&L for management review.

8. Cross-Module Integration

One of SAP's most powerful features is seamless automatic integration between modules. When a business event occurs in MM or SD, a corresponding FI accounting document is created automatically — eliminating manual entries and ensuring consistency. The following table summarizes all key integration points for TechNova:

Business Event	Trigger Module	Automatic FI Posting	GL Accounts
Goods Receipt (PO)	MM (MIGO)	Inventory Dr, GR/IR Cr	130100 / 200200
Invoice Verification	MM (MIRO)	GR/IR Dr, AP Cr, GST Input Dr	200200 / 200100 / 160300
Goods Issue (Production)	MM (MIGO)	COGS Dr, Raw Material Inventory Cr	500100 / 130100

Business Event	Trigger Module	Automatic FI Posting	GL Accounts
Post Goods Issue (Delivery)	SD (VL02N)	COGS Dr, Finished Goods Inventory Cr	500100 / 130300
Customer Billing	SD (VF01)	AR Dr, Revenue Cr, GST Output Cr	100300 / 400100 / 210300
Customer Payment	FI (F-28)	Bank Dr, AR Cr	100200 / 100300
Vendor Payment	FI (F110)	AP Dr, Bank Cr	200100 / 100200

9. Unique Features of This Blueprint

Feature	Description
Indian FY & GST	Blueprint uses Fiscal Year Variant V3 (Apr–Mar) matching Indian financial year, with full GST configuration (CGST, SGST, IGST) using procedure TAXIN.
Three-Module Coverage	Unlike single-module implementations, this blueprint integrates FI, MM, and SD with documented automatic account determination (OBYC, VKOA).
Real Master Data	35 GL accounts, 20 materials (ROH/HALB/FERT/HIBE), 15 vendors, 15 customers — all with realistic Indian business data, GSTINs, and pricing.
Release Strategy	Multi-level approval workflows for PR and PO based on value thresholds — a real-world control mechanism often missing in academic blueprints.
S/4HANA Readiness	The entire blueprint is designed to be compatible with SAP S/4HANA — using universal journal concepts and simplified data models where applicable.
Transaction Log Files	Sample P2P and O2C transaction logs provided as Excel workbooks demonstrating document numbers, status tracking, and value flows.

10. Results

The SAP ERP blueprint for TechNova Manufacturing Pvt. Ltd. yielded the following outcomes:

- **Organizational Structure:** A complete SAP enterprise structure was defined across FI, MM, and SD — covering 27 organizational units from Client level down to Storage Locations and Sales Areas.

- **Configuration Completeness:** Step-by-step configuration checklists were documented for all three modules, covering 10 phases in FI, 6 phases in MM, and 8 phases in SD with over 60 unique SAP transaction codes.
- **Master Data Depth:** A realistic master data set was created including 35 GL accounts (segregated into 8 account groups), 20 materials across 4 material types, 15 vendors across 3 account groups, and 15 customers across 3 account groups.
- **Process Documentation:** Three complete end-to-end business processes (P2P, O2C, R2R) were documented with step-by-step instructions, T-codes, FI postings, document flows, and KPI targets.
- **Integration Design:** All FI-MM and FI-SD integration touchpoints were mapped with specific GL accounts, posting keys, and OBYC/VKOA configuration references — providing a complete accounting picture of each business event.

11. Future Scope

The current blueprint represents a solid foundation for a Phase 1 SAP implementation. The following enhancements would significantly increase the scope and value of the system in subsequent phases:

- **SAP S/4HANA Migration:** Upgrading from SAP ECC 6.0 to S/4HANA would enable real-time analytics through SAP HANA's in-memory processing, the Universal Journal (table ACDOCA), and simplified logistics.
- **SAP PP (Production Planning):** Adding the PP module would enable the complete Plan-to-Produce cycle — linking material requirements planning (MRP) to production orders, capacity planning, and shop floor control.
- **SAP HR / SuccessFactors:** Extending to Human Resources would enable the Hire-to-Retire process including payroll processing, which currently has salary as a manual journal entry.
- **Power BI / SAP Analytics Cloud Integration:** Connecting SAP data to Microsoft Power BI or SAP Analytics Cloud would enable interactive procurement and sales dashboards for management reporting.
- **Vendor Portal (SAP Ariba):** Integrating SAP Ariba for upstream sourcing, e-auctions, and supplier onboarding would extend the P2P cycle into strategic sourcing territory.
- **GST e-Invoicing (IRP Integration):** Implementing SAP's GST e-Invoicing framework to auto-generate IRN (Invoice Reference Numbers) and QR codes for invoices above the statutory threshold.

12. Conclusion

This capstone project has successfully designed and documented a comprehensive SAP ERP implementation blueprint for TechNova Manufacturing Pvt. Ltd. By defining the complete organizational structure, module configuration, master data, and end-to-end business processes, the project demonstrates how SAP enables structured, transparent, and auditable operations across procurement, sales, and finance functions.

The integration between SAP FI, MM, and SD — where every goods movement, goods issue, and customer billing automatically generates a corresponding accounting document — is the core value proposition of an ERP system. This blueprint captures that integration in full, providing a reference that could guide an actual SAP implementation for a company of similar profile.

In conclusion, a well-implemented SAP system is not merely an operational tool — it is a strategic asset. When the three modules work together with correctly configured master data and business processes, TechNova would achieve full financial control, real-time inventory visibility, and end-to-end sales traceability from a single unified platform.

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